

Question 1

Given the following piece of code of C language:

```
#include <stdio.h>
int main(void) {
    int x=243;

    int y = (x>0 || x++/3);

    printf("%d\n", x);

    return 0;
}
```

What is the output ?

Question 2

Given the following **Python** function

```
def fun1(a, b, c=2, d=1):
    return a-c+(b+d)
```

What are the outputs of the following function call:

```
fun1(79,1,55)
```

Question 3

Given the following piece of code of C language:

```
int c= 42%7;
```

what is the value of c?

Question 4

Given the following piece of code of **Python**:

```
a=141
```

```
b=2
```

```
a,b=b,a
```

what is the value of a?

Question 5

Given the following **Python** function

```
def fun1(a, b=5, c=2, d=1):  
    return a-c+(b+d)
```

What are the outputs of the following function call:

fun1(48)

Question 6

What is the value of the following C++ expression:

int(30.842)

Question 7

What is the value of the following **Matlab** expression:

4^3^2

Question 8

0.75 points

Save Answer

Find the address of the element A(6,3) where A is a two-dimensional array with subscript limits *number of columns = 10* and *number of rows = 13* and A is in row-major addressing. Assume base address is **1000** and word size is **2** and lower bound for both row and column is 1.

Question 9

Given the following piece of code of **Perl** language:

```
$c= -33%19;
```

what is the value of \$c?

Question 10

What is the value of x in the following Python expression:

```
x = 15 if 27>43 else 37
```

Question 11

Given the following piece of code of C language:

```
float f = 84/9;
```

what is the value of f?

Question 12

What is the value of x in the following **C** code

```
int a = 52;
```

```
int x = !a ;
```

Question 13

What is the value of a of the following **Perl** language?

```
#!/usr/local/bin/perl
```

```
$a = 18;
```

```
until( $a < 70 ){  
    $a = $a + 1;
```

```
}
```

Question 14

What is the value of the variable **result** in the following **Python** code?

```
sum=0
count =1
result = 2

if sum == 0 :
    if count == 0 :
        result = 0
    else:
        result = 1
```

Question 15

Given the following piece of code of C language:

```
int c= -69%7;
```

what is the value of c?

Question 16

Given the following piece of code of C language:

```
#include <stdio.h>
int main(void) {
    int x=110;

    int y = (-9 && x++/3);

    printf("%d\n", x);

    return 0;
}
```

What is the output ?

Question 17

What is the value of the following C expression:

$42 > 92 > 64$

Question 18

Given the following piece of code of C language:

```
int x= 13;  
if (x=54)  
printf("%d",x++);  
else  
printf("%d", x--);
```

What is the output?

Question 19

What is the value of the following Haskell expression:

```
(- (* (+ 6 22) 2) 11)
```

Question 20

0.75 points

Save Answer

Find the address of the last element of array A, where A is a two-dimensional array with subscript limits *number of columns* = 7 and *number of rows* = 9 and A is *in row-major* addressing. Assume base address is 1000 and word size is 2 and lower bound for both row and column is 1.

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Question 21

What is the value of the following APL expression:

```
34*64+1
```


Question 22

What is the value of the following **Python** expression:

```
8**0**4
```

Question 23

Given the following **Python** code, what is the value of A[5\]?

```
A= [X *Y for X in range (1,5) for Y in range (2,7) if Y < 4\]
```

Question 24

What is the value of x in the following Pascal expression:

```
x := 5 div 4;
```

Question 25

Given the following piece of code of C language:

```
#include <stdio.h>
int main(void) {
    int x=26;

    int y = (x=0 && x++/10);

    printf("%d\n", x);

    return 0;
}
```

What is the output ?

Question 26

What is the value of x in the following **Python** expression:

```
x = 17 // 2
```

Question 27

In **Matlab**, suppose we have an array x

```
2 4 26 86
3 17 16 39
0 1 6 30
7 31 0 6
```

what are the value of x(8)